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Original Contribution

Impact of ultrasound-guided erector spinae plane block on postoperative quality of recovery in video-assisted thoracic surgery: A prospective, randomized, controlled trial

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ABSTRACT

Study objective: Regional anesthesia improves postoperative analgesia and enhances the quality of recovery (QoR) after surgery. We examine the efficacy of ultrasound-guided erector spinae plane block (ESPB) on QoR after video-assisted thoracic surgery (VATS).

Design: Prospective, randomized, double-blinded, placebo-controlled trial.

Setting: Single institution, tertiary university hospital.

Patients: Adult patients who scheduled for VATS under general anesthesia were enrolled in the study.

Interventions: We randomly allocated patients to receive preoperative ultrasound-guided ESPB with 25 ml of either 0.5% ropivacaine (ESPB group) or normal saline (Control group).

Measurements: The primary outcome was QoR as measured by the 40-item QoR questionnaire (QoR-40) score at postoperative day 1. Secondary results were post-anesthesia care unit (PACU) discharge time, acute postoperative pain, cumulative opioid consumption, the incidence of postoperative nausea or vomiting (PONV), and patient satisfaction.

Main results: The global QoR-40 score at postoperative day 1 (median, interquartile range) was significantly higher in the ESPB group (174, 170 to 177) than the control group (161.5, 160 to 165), estimated median difference 11 (95% CI 9 to 13, $P < 0.001$). Compared with the control group, single-injection of ESPB reduced PACU discharge time, acute postoperative pain, and cumulative opioid consumption. Correspondingly, the median patient satisfaction scores were higher in the ESPB group than the control group (9 versus 7, $P < 0.001$).

Conclusion: Preoperative single-injection thoracic ESPB with ropivacaine improves QoR, postoperative analgesia, and patient satisfaction after VATS.

1. Introduction

Video-assisted thoracic surgery (VATS) is a minimally invasive technique that allows faster recovery after thoracic surgery. Although acute pain after VATS is less than an open thoracotomy, postoperative analgesia remains challenging [1]. Patients scheduled for VATS experience moderate to severe postoperative pain, which may result in an increased risk of postoperative complications, negatively affect functional recovery, and increased the risk of persistent postsurgical pain [2]. Therefore, adequate postoperative pain relief is imperative to improve functional outcomes and accelerate discharge from the hospital [3]. Local-regional analgesia is a core component of multimodal analgesia for postsurgical pain. However, the optimal analgesic technique for VATS procedures has not been well established [4]. After all, thoracic epidural analgesia and paravertebral block are accompanied by severe but rare complications, e.g., epidural hematoma, pneumothorax, and total spinal anesthesia [5,6].

Erector spinae plane block (ESPB) is a new ultrasound-guided

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regional block where a local anesthetic is injected into the interfacial plane between the transverse spinal processes and the erector spinae muscle [7]. Given its safety, ease of performance, and efficacy, ESPB could be an attractive alternative strategy for analgesia after VATS [8]. Several clinical studies have reported that ESPB could provide analgesia in many types of surgeries, including thoracoscopic surgery [9–11]. However, to date, the effectiveness of ESPB on the quality of recovery (QoR) following VATS has not yet been evaluated.

Therefore, we conducted this prospective randomized controlled study to examine the hypothesis that the inclusion of ESPB to a multimodal analgesic regimen would enhance patient-perceived QoR after VATS, according to the Consolidated Standards of Reporting Trials (CONSORT) 2010 statement [12].

2. Material & method

2.1. Study design

Ethical approval for this study (Identifier: K2018-10-001) was provided by the Ethical Committees of Fujian Provincial Hospital, Fujian, China (Chairperson *Prof.* Fayang Lian) on October 16th, 2018. We registered the study protocol at ClinicalTrials.gov (Identifier: NCT03756987) on November 28th, 2018. Adult American Society of Anesthesiologists (ASA) physical status I or II patients, aged 18 to 65 years, scheduled for elective VATS between December 2018 and November 2019 were eligible for trial inclusion. Patients with a history of allergy to local anesthetics or non-steroidal anti-inflammatory drugs, known coagulation disorders, infection near the puncture site, a history of chronic pain or regular analgesic use, inability to communicate, or other reasons that were not appropriate for this study were excluded.

After obtained informed consent, all individual participants included in the study were randomly allocated to receive ESPB with 25 ml of study drug, either ropivacaine 0.5% (ESPB group) or normal saline (Control group). Randomization was carried out on a 1:1 ratio using a computer-generated randomization sequence by an assistant not involved in this study. We concealed allocations in numbered opaque envelopes. A research nurse who was not involved in the study opened the envelopes and prepared the study drug on the day of surgery. Consequently, the patients, attending anesthesiologists, surgeons, PACU personnel, data collectors, and the person who performed the statistical analysis, were blinded to the group assignment throughout the whole observation period, including all postoperative follow-ups.

2.2. Ultrasound-guided erector spinae plane block

Upon arrival in the operating room, all patients were administered midazolam 2 mg intravenously as premedication before ultrasound-guided ESPB. After placing the patient in the lateral position, a single experienced attending anesthesiologist performed all unilateral ESPB before general anesthesia. By palpation of spinous processes starting from the C7 downward, the T5 spinous process was identified. Then, a 6–13 MHz linear array transducer (HFL38x, FUJIFILM SonoSite, USA) was located longitudinally 2 to 3 cm lateral to the T5 transverse process. After achieving clear visualization of the trapezius, rhomboid major, and erector spinae muscles superficial to the T5 transverse process, we inserted an 80-mm 22-gauge needle (Stimuplex D, Braun, Germany) in a caudal-to-cephalad direction until the needle tip lay in the plane deep to the erector spinae muscles by using an in-plane technique. Once the correct location of the tip was confirmed by injecting 2–3 ml of saline solution, 25 ml of 0.5% ropivacaine or saline was injected into the target interfacial plane.

2.3. Standard general anesthesia and postoperative analgesia protocol

The anesthesia technique was standardized. We induced general anesthesia by using a bolus injection of propofol 2 mg kg^{-1}, sufentanil 0.6 μg kg^{-1}, and rocuronium 0.6 mg kg^{-1}. The position of the double-lumen tracheal tube was corrected using bronchoscopy, and the patient was placed in a lateral decubitus position. All patients were administered dexamethasone 10 mg and tropisetron 5 mg intravenously for nausea and vomiting prophylaxis immediately after tracheal intubation. General anesthesia was maintained by inhalation of sevoflurane and remifentanil infusion titrated to maintain the hemodynamic parameters fluctuation within 20% of baseline and a bispectral index range of 40 to 50. Muscle relaxation was achieved by intermittent injections of rocuronium as needed. At the end of the procedure, neuromuscular blockade was antagonized by using neostigmine 40 μg kg^{-1} and atropine 20 μg kg^{-1}. After the trachea was extubated, all patients were transferred to the PACU for observation at least 2 h. For postoperative analgesia, all patients received patient-controlled intravenous analgesia (PCIA) with sufentanil. The PCIA pump (REHN11, Renxian Medical Corporation, Jiangsu, China) was set to deliver a background infusion of sufentanil 2 μg h^{-1}. If the numeric rating scale (NRS) for pain exceeded three or the patient required, a bolus injection of sufentanil 2 μg was administrated as a rescue analgesic, with a 10-min lockout interval via the PCIA pump. Patients were also prescribed regular analgesia, namely intravenous injection parecoxib sodium (a non-steroidal anti-inflammatory drug) 40 mg every 12 h.

2.4. Outcome measures

The primary outcome was postoperative QoR after VATS, which was assessed by using the Chinese version of the global QoR-40 score at postoperative day (POD) 1. The QoR-40 incorporates 40 questions that assess five domains of QoR, namely emotional status, physical comfort, psychological support, physical independence, and pain. Each question is graded on a five-point Likert scale. The global QoR-40 score ranges from 40 to 200, representing very poor to excellent [13]. Secondary outcomes included PACU discharge time, acute postoperative pain, postoperative cumulative opioid consumption, the incidence of PONV, ESPB-related adverse events, and patient satisfaction. A single trained research nurse blinded to group assignment assessed the above outcomes. The PACU discharge time was defined as the time from PACU admission to the modified Aldrete score reached 10 [14]. Postoperative pain intensity at rest and during coughing was recorded by using an NRS score at 0.5, 1, 2, 4, 8, 24, and 48 h after surgery. Patient satisfaction was assessed at POD 2 by using an 11-point Likert scale (0 = entirely unsatisfied; 10 = fully satisfied) [15].

2.5. Sample size and statistical analyses

Our sample size was calculated based on the global QoR-40 score. A 10-point difference in the global QoR-40 score represents a clinical improvement in QoR after surgery and anesthesia [16,17]. According to our preliminary study, the global QoR-40 score at POD1 was 164 (14.6). Assuming alpha = 0.05, beta = 0.2 for a 10-point difference in QoR-40 scores between groups, the calculated sample size was 34 patients per group. To allow for a possible 10% dropout rate, we enrolled a total of 76 participants in this study.

We assessed the normality of quantitative variables for normality by using the Shapiro-Wilk test and Q-Q plots. Normally distributed variables were reported as mean ± standard deviation (SD) and analyzed by using independent samples *t*-test. Non-normally distributed variables were reported as median (interquartile range, IQR) and analyzed by using Mann-Whitney *U* test. Categorical variables were summarized as number (percentage) and analyzed by using Fisher's exact or χ^2 test as appropriate. Besides, the NRS pain scores were evaluated by using repeated-measures analysis of variance, and post hot multiple comparisons were performed with Bonferroni correction. All statistical analyses were performed using SPSS 25.0 (IBM Corporation, NY, USA). We considered $P < 0.05$ (two-tailed) statistically significant.

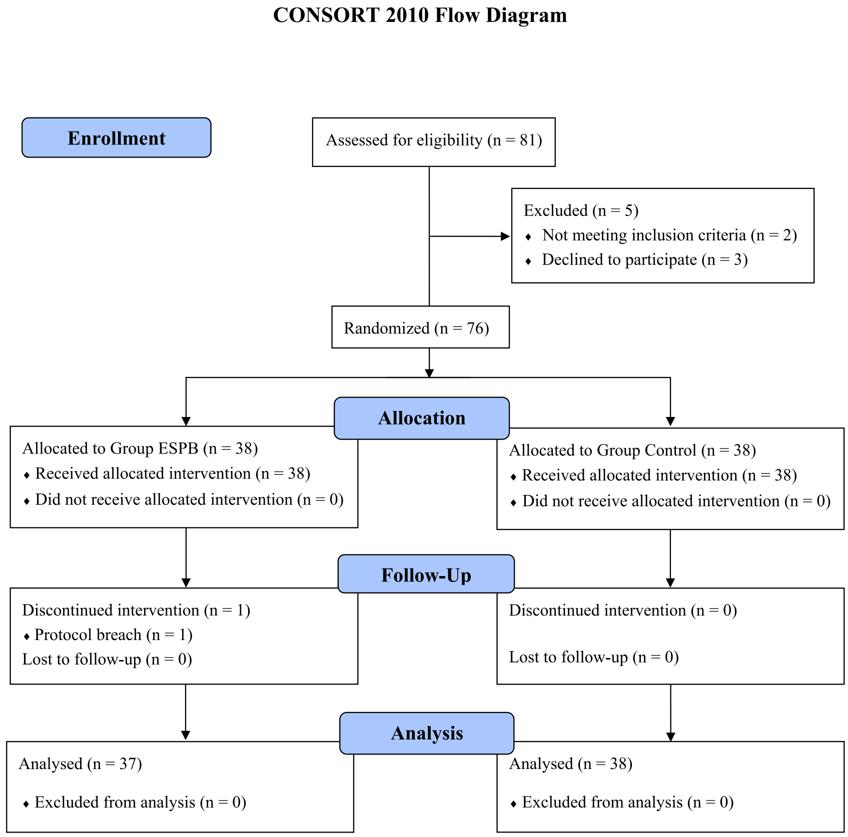

Fig. 1. Consolidated Standards of Reporting Trials (CONSORT) flow diagram depicting the progress of patients through the study. Abbreviation: ESPB, erector spinae plane block.

3. Results

The flowchart is detailed in Fig. 1, in accordance with the CONSORT 2010 statement [12]. Of the 81 patients assessed for eligibility, two were deemed ineligible based on the inclusion criteria, and three refused to participate. In the ESPB group, one patient was excluded from the analysis because of protocol breach (change to open thoracotomy). Consequently, a total of 75 patients were analyzed in the study. Patient demographics and operation characteristics, including preoperative global QoR-40 scores, were similar between the two groups (Table 1).

The global QoR-40 scores are presented in Fig. 2. At POD1, the global QoR-40 score was higher in the ESPB group compared with the control group (median 174, IQR 170–177, vs. median 161.5, IQR 160–165; $P < 0.001$), with a median difference of 11 (95% CI 9 to 13, $P < 0.001$). At POD2, the global QoR-40 score was higher in the ESPB group compared with the control group (median 181, IQR 177–184, vs. median 169.5, IQR 166–172; $P < 0.001$), with a median difference of 10 (95% CI 9 to 13, $P < 0.001$).

The postoperative NRS pain scores are presented in Table 2. Briefly, preoperative ESPB reduced NRS pain scores both at rest and during coughing in the first postoperative 8 h (all $P < 0.001$). However, at 24 and 48 h postoperatively, there was no difference between the groups in regards to NRS pain scores at rest ($P = 0.17$ and $P = 0.109$, respectively), or during coughing ($P = 0.173$ and $P = 0.159$, respectively). Similarly, postoperative cumulative sufentanil consumption in the first postoperative 24 h was lower in the ESPB group (median 50 μg, IQR 48–54, vs. median 68 μg, IQR 66–72; $P < 0.001$), with a median difference of −18 μg (95% CI −20 to −16, $P < 0.001$).

As detailed in Table 3, PACU discharge time in the ESPB group was earlier than that in the control group (estimated median difference −20 min, 95% CI −20 to −15, $P < 0.001$). The incidence of PONV was 2/37 (5.4%) in the ESPB group and 7/38 (18.4%) in the control group, which did not differ significantly between the groups (relative risk 0.253, 95% CI 0.049 to 1.31, $P = 0.153$). Besides, patient satisfaction scores were higher in the ESPB group compared with the control group (median 9, IQR 8–9, versus median 7, IQR 6–8; $P < 0.001$). No patient experienced ESPB-related adverse events, including pneumothorax, bleeding, or local anesthetic toxicity during the study.

Table 1
Patient demographics and operation characteristics.

Variables	Group ESPB (n = 37)	Group control (n = 38)	*P*-value
Age, years	56.0 (50.5–62.0)	58.0 (52.0–61.0)	0.656
Height, cm	164 (158–171)	165 (158–172)	0.51
Weight, kg	55.0 (51.5–64.5)	60.0 (54.3–68.0)	0.183
Gender, n (%)			0.884
Male	14 (37.8)	15 (39.5)	
Female	23 (62.2)	23 (60.5)	
ASA physical status, n (%)			0.711
I	5 (13.5)	3 (7.9)	
II	32 (86.5)	35 (92.1)	
Type of surgery, n (%)			0.516
Lobectomy	31 (83.8)	34 (89.5)	
Segmentectomy	6 (16.2)	4 (10.5)	
Duration of surgery, min	180 (175–190)	185 (177–198)	0.09
Preoperative global QoR-40 score	183 (178.5–186)	183.5 (180–186)	0.259

Notes: Data are presented as median (interquartile range) or number (percentage).
Abbreviations: ESPB, erector spinae plane block; ASA, American Society of Anesthesiologists; QoR-40, 40-item quality of recovery questionnaire.

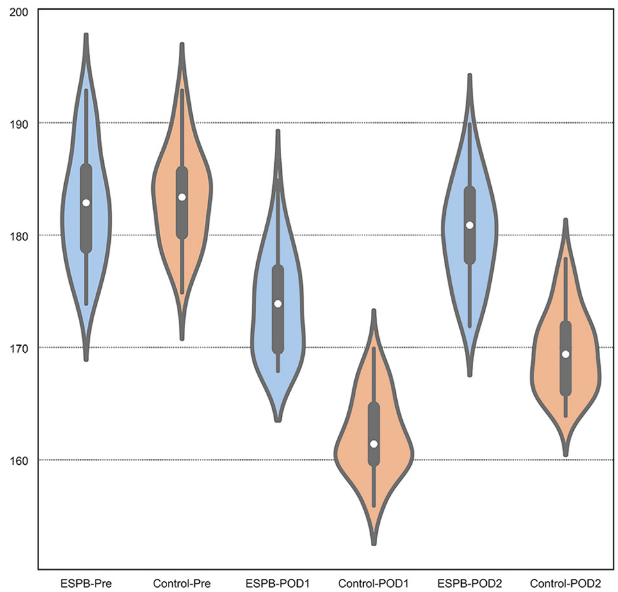

Fig. 2. Patients in the erector spinae plane block (ESPB) group had higher global Quality of Recovery (QoR)-40 scores than the control group at postoperative day (POD) 1 and 2. Notes: The violin plots visualize a distribution of quantitative data as a continuous approximation of the probability density function by using kernel density estimation. The densities are additionally annotated with the median value (white dot) and interquartile range (black line).

Table 2
Postoperative numerical rating scale (NRS) pain scores.

	Group ESPB (n = 37)	Group control (n = 38)	P value
NRS at rest			
0.5 h	1 (0–1)	3 (2–3)	< 0.001
1 h	1 (0.5–1)	3 (2–3)	< 0.001
2 h	1 (1–2)	3 (2–3)	< 0.001
4 h	2 (1–2)	3 (2–3)	< 0.001
8 h	2 (1–2)	3 (2–3)	< 0.001
24 h	2 (2–2)	2 (2–2)	0.17
48 h	2 (1–2)	2 (2–2)	0.109
NRS during coughing			
0.5 h	2 (1–2)	3 (3–4)	< 0.001
1 h	2 (2–3)	5 (4–5)	< 0.001
2 h	2 (2–3)	3.5 (3–4)	< 0.001
4 h	2 (2–2.5)	3 (3–4)	< 0.001
8 h	2 (2–3)	3 (3–4)	< 0.001
24 h	3 (2–3)	3 (3–3)	0.173
48 h	3 (2–3)	3 (2–3)	0.159

Notes: Data are presented as median (interquartile range). Mann–Whitney *U* test was used to compare medians between the two groups.
Abbreviations: ESPB, erector spinae plane block.

4. Discussion

We demonstrated that patients who received preoperative ESPB reported higher global QoR-40 scores at PODs 1 and 2, indicating a higher quality of recovery after VATS. Overall, a single injection of ESPB has been associated with early discharge from the PACU, provided superior pain relief in the early postoperative period, and reduced the postoperative cumulative opioid consumption. Furthermore, we demonstrated that patients exposed to preoperative ESPB had increased satisfaction with their pain management. These findings suggested that the preoperative single injection of ESPB, as part of the multimodal analgesic regimen, may be an effective intervention for enhanced recovery after VATS.

We chose the QoR-40 score rather than the pain score or cumulative opioid consumption as the primary outcome. The Standardized Endpoints in Perioperative Medicine initiative states that one or more of six recommended endpoints should be considered in clinical trials assessing patient comfort after surgery [18]. One of them is the QoR-40 score, a validated patient-centered multidimensional quality assessment tool [19]. Recently, Myles et al. point out a 6.3-point difference or more in the global QoR-40 score represents a clinically relevant improvement in the quality of recovery after surgery [20]. In this study, we demonstrated that the preoperative single injection of ESPB with ropivacaine resulted in increased 11 points of the global QoR-40 scores at POD1. Therefore, our results indicate that preoperative ESPB leads to the significantly better postoperative health status of patients after VATS, which was in agreement with previous studies [21,22].

Table 3
Secondary Outcomes during the study period.

	Group ESPB (n = 37)	Group control (n = 38)	P value
Remifentanil dose, μg kg^{-1} min^{-1}	0.11 (0.10–0.12)	0.14 (0.13–0.16)	< 0.001
PACU discharge time, min	30 (25–35)	45 (40–50)	< 0.001
Cumulative sufentanil consumption, μg			
0–24 h	50 (48–54)	68 (66–72)	< 0.001
24–48 h	58 (55–61)	60 (56–63)	0.327
PONV, n (%)	2 (5.4)	7 (18.4)	0.153
ESPB-related adverse events, n (%)	0	0	NS
Patient satisfaction score	9 (8–9)	7 (6–8)	< 0.001

Notes: Data are presented as median (interquartile range) or number (percentage).
Abbreviations: ESPB, erector spinae plane block; PACU, post-anesthesia care unit; PONV, postoperative nausea or vomiting.

Another important finding of this study is the opioid-sparing properties and better analgesia obtained by preoperative ESPB. In this study, patients in the ESPB group exhibited significantly lower pain scores during the first 8 h after surgery. But, the statistical differences did not persist beyond 24 h, which may occur due to the pharmacokinetic properties of ropivacaine. The use of a continuous local anesthetic infusion to complement single-injection ESPB may provide prolonged postoperative analgesia and address this issue. Similar to multiple studies [23–25], a single injection of ESPB also reduced opioid consumption during the immediate postoperative period. Opioid-related side effects are numerous, such as respiratory depression, nausea and vomit, and bowel ileus. Therefore, whenever possible, the use of opioids for analgesia should be avoided or at least reduced to enhance recovery. Nevertheless, ESPB could be a promising component of a multimodal analgesia approach in VATS.

In addition to the beneficial effect of ESPB in the overall quality of recovery and opioid-sparing, patients in the ESPB group had a higher satisfaction score compared with the control group. Most likely, prolonged time that ESPB provided pain relief, lower pain scores at rest and during coughing, decreased cumulative sufentanil consumption, and reduced postoperative nausea and vomiting were all factors that resulted in higher patient satisfaction with their pain management. But, we failed to demonstrate any significant differences in terms of PONV between groups. Furthermore, no ESPB-related complications such as pneumothorax, injury to vascular or nervous structures, and local anesthetic toxicity were identified in the study. One possibility might be that the research was not adequately powered to detect differences in these secondary measures.

Despite ESPB has recently become a popular regional anesthesia technique, there is still controversy regarding the mechanism of ESPB [26–28]. Both cadaveric and clinical reports have shown the approach to ESPB may be the most crucial factor for block success and local anesthetic spread [8,29]. There are already many modifications of ESPB in literature. Recently, for example, Coşarcan et al. [30] reported that local anesthetic was administered above superior costotransverse ligament resulted in better clinical results.

There are several limitations of this study should be noted. First, we did not assess the distribution of sensory blockade and detect potential block failures so as not to increase the risk of unblinding the group assignment. Therefore, we could not explore an association between ESPB and analgesic outcomes, including postoperative pain intensity and opioid consumption. Second, the generalizability of the results is limited, considering that the present study lacked a typical multimodal analgesic regimen, such as wound infiltration and basal infusion free PCIA. Third, the sample size was calculated based on the primary outcome. Therefore, it is possible that the current study was underpowered and subject to type-II error in terms of secondary outcome metrics. Fourth, the Chinese version of the QoR-40 questionnaire was used in the present study, which has not been formally validated. Consequently, language and cultural differences may have distorted the results of this study. Lastly, we have failed to compare thoracic ESPB with thoracic paravertebral blocks (active control intervention) in the present study. Further studies are required to address these limitations.

5. Conclusion

In patients undergoing VATS, a preoperative single injection of ESPB with ropivacaine is associated with improved quality of recovery and postoperative analgesia. Given the ease of performance and the theoretical safety profile, we suggest the incorporation of ESPB into the multimodal postoperative analgesia protocol in VATS.

CRediT authorship contribution statement

Yusheng Yao: Conceptualization, Methodology, Writing - review & editing, Funding acquisition. **Shiwei Fu:** Formal analysis, Visualization, Writing - original draft. **Shuanbo Dai:** Investigation, Resources, Project administration. **Jia Yun:** Methodology, Investigation. **Minghui Zeng:** Data curation, Writing - original draft. **Hao Li:** Conceptualization, Project administration, Funding acquisition. **Xiaochun Zheng:** Conceptualization, Validation, Writing - review & editing, Supervision, Project administration, Resources, Funding acquisition.

Declaration of competing interest

The author reports no conflicts of interest in this work.

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